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Communication fabric structures for increased bandwidth

Abstract

An apparatus includes first agents configured to transfer transactions using an ordered protocol, as well as second agents configured to transfer transactions using a protocol with no enforced ordering. The apparatus may also include input/output (I/O) interfaces coupled to respective ones of the first agents and configured to enforce the ordered protocol. The apparatus may further include a communication network including a plurality of network switches. A particular one of the network switches may be coupled to at least one other network switch of the plurality. The apparatus may also include a network interface coupled to the second agents, to the I/O interfaces, and to the particular network switch. This network interface may be configured to transfer data transactions between the second agents and the particular network switch, and to transfer data transactions between the I/O interfaces and the particular network switch.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6973079	12/2004	Carlson	370/395.71	H04L 49/30
7437518	12/2007	Tsien	N/A	N/A
7529245	12/2008	Muller	370/395.7	H04L 69/324
9119109	12/2014	Dubrovsky et al.	N/A	N/A
2008/0232364	12/2007	Beverly	N/A	N/A
2009/0070775	12/2008	Riley	719/321	H04L 49/65
2013/0166812	12/2012	Boucard	710/314	G06F 13/405
2014/0114928	12/2013	Beers et al.	N/A	N/A
2014/0365632	12/2013	Ishii et al.	N/A	N/A
2015/0006749	12/2014	Hendel et al.	N/A	N/A
2016/0065484	12/2015	Suzuki	370/415	H04L 47/623
2018/0331960	12/2017	Browne	N/A	H04L 47/2441
2019/0312772	12/2018	Zhao	N/A	H04L 41/12
2020/0213217	12/2019	Nye	N/A	G06N 3/084
2022/0232111	12/2021	Ford	N/A	H04L 49/9036
2022/0311544	12/2021	Alverson et al.	N/A	N/A
2023/0053530	12/2022	Hammarlund et al.	N/A	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion in PCT Appl. No. PCT/US2024/043732 mailed Dec. 2, 2024, 13 pages. cited by applicant

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Background/Summary

(1) The present application claims priority from U.S. Provisional App. No. 63/583,899, entitled “Communication Fabric Structures for Increased Bandwidth,” filed Sep. 20, 2023, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

Technical Field

(1) Embodiments described herein are related to computing systems including, for example, systems-on-a-chip (SoCs). More particularly, embodiments are disclosed relating to techniques for increasing bandwidth through a communication fabric.

Description of the Related Art

(2) A network fabric interconnect may provide high bandwidth and low latency transport layers between various agents coupled across a plurality of networks in an integrated circuit or multichip system. Such interconnect architectures may be designed to have various specialized lanes for transporting data between each of the various agents, for example, central processing units (CPUs), graphic processing units (GPUs), neural processing engines, memory systems and the like. To support a unified memory space, a high bandwidth network fabric may employ network switches that are fully buffered. Various types of peripheral circuits (e.g., “agents”) may be included in these systems and may employ a variety of communication protocols. These agents may be coupled to ones of the network switches via one or more forms of networking interfaces.

(3) In some systems, a communication fabric may include each agent being connected to one of several network interfaces which, in turn, may be coupled to the communication fabric. Such a technique may have a high communication latency, multiple protocol conversions, a high-level of data buffering, and a high-level of power consumption. In addition, a protocol with ordering rules may be applied for agents that do not require an ordered protocol, thereby further reducing performance in some cases.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following detailed description makes reference to the accompanying drawings, which are now briefly described.

(2) FIG. **1** illustrates a block diagram of an embodiment of a system that includes a communication network coupled to a plurality of agent circuits via network interfaces and input/output (I/O) interfaces.

(3) FIG. **2** shows a block diagram of a portion of an embodiment of the system of FIG. **1** in which a plurality of protocols is used.

(4) FIG. **3** depicts a block diagram of a portion of an embodiment of the system of FIG. **1** in which an ordered protocol is enforced for some agent circuits.

(5) FIG. **4** illustrates a block diagram of another embodiment of a system that includes a communication network coupled to a plurality of agent circuits via network interfaces and I/O interfaces.

(6) FIG. **5** shows a flow diagram of an embodiment of a method for transferring data transactions using network interfaces and I/O interfaces.

(7) FIG. **6** shows a flow diagram of an embodiment of a method for transferring responses to data transactions using network interfaces and I/O interfaces.

(8) FIG. **7** depicts a block diagram of an embodiment of a system with a communication network with multiple communication lanes.

(9) FIG. **8** illustrates a block diagram of an embodiment of a system with two SoCs, each with a communication network with multiple communication lanes.

(10) FIG. **9** shows a block diagram of an embodiment of a network with a ring topology.
(11) FIG. **10** depicts a block diagram of an embodiment of a network with a mesh topology.
(12) FIG. **11** illustrates a block diagram of an embodiment of a system that includes a network interface with asymmetrical input and output channels.
(13) FIG. **12** shows a flow diagram of an embodiment of a method for transferring data transactions using a network interface with asymmetrical input and output channels.
(14) FIG. **13** depicts various embodiments of systems that include integrated circuits that utilize the disclosed techniques.
(15) FIG. **14** is a block diagram of an example computer-readable medium, according to some embodiments.
(16) While embodiments described in this disclosure may be susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the embodiments to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the appended claims.

DETAILED DESCRIPTION OF EMBODIMENTS

(17) Various integrated circuits and multichip systems may employ a plurality of communication networks. As used herein, "communication network," or simply "network," refers collectively to various agents that communicate, via a common set of network switches. Such networks may be physically independent (e.g., having dedicated wires and other circuitry that form the network) and logically independent (e.g., communications sourced by agents in the system may be logically defined to be transmitted on a selected network of the plurality of networks and may not be impacted by transmission on other networks). In some embodiments, network switches may be included to transmit packets on a given network. As used herein, an "agent" refers to a functional circuit that is capable of initiating (sourcing) or being a destination for communications on a network. An agent may generally be any circuit (e.g., CPU, GPU, neural processing engine, peripheral, memory controller, etc.) that may source and/or receive communications on a given network. A source agent generates (sources) a communication, and a destination agent receives the communication. A given agent may be a source agent for some communications and a destination agent for other communications. In some cases, communication between two agents (also referred to as a "transaction") may cross between two or more of the networks.
(18) By providing physically and logically independent networks, high bandwidth may be achieved via parallel communication on the different networks. Additionally, different traffic may be transmitted on different networks, and thus a given network may be optimized for a given type of traffic. For example, a multicore CPU in an system may be sensitive to memory latency and may cache data that is expected to be coherent among the cores and memory. Accordingly, a CPU network may be provided on which the cores and the memory controllers in a system are agents. Another network may be an input/output (I/O) network. This I/O network may be used by various peripheral devices ("peripherals") to communicate with memory. The network may support the bandwidth needed by the peripherals and may also support cache coherency. Furthermore, the system may additionally include a relaxed order network. The relaxed order network may be non-coherent and may not enforce as many ordering constraints as an I/O or a CPU network. The relaxed order network may be used by GPUs to communicate with memory controllers. Other embodiments may employ any subset of the above networks and/or any additional networks, as desired.
(19) This combination of networks in a system may be referred to as a "communication fabric," a "network fabric," or simply a "fabric." In some instances, a "global fabric" may be used to refer to the various communication paths that are "woven" across all networks in a system. A "local fabric," therefore, may refer to communication paths "woven" across a subset of networks and or

portions of a network. As described above, a communication fabric may include a number of agents being connected to a network interface which, in turn, may be coupled to the communication fabric. In some embodiments, a group of agents may be coupled to an input/output (I/O) interface that resides between the network interface and the group of agents.
(20) Use of an I/O interface between agents and network interfaces may allow for use of intellectual property (IP) circuit designs from variety of IP providers by accommodating multiple data formats and communication protocols. An I/O interface may support translations from one or more data formats to a data format supported by the network interface. In addition, such an I/O interface may support an ordered transaction protocol in which transactions are sent to the network interface in an order in which they are received from an agent. Any responses to an ordered set of transactions may also be returned to the source agent in an order in which the source agent sent the transactions.
(21) Use of an I/O interface between agents and network interfaces may also result in high communication latency, multiple protocol conversions, a high-level of data buffering, thereby resulting in a high-level of power consumption. In addition, application of ordering rules may be applied for agents that do not require an ordered protocol, thereby further reducing performance in some cases. Accordingly, a reduced I/O interface is proposed for use with peripheral agents that require ordered protocols. A multiport network interface is also proposed to communicate directly with a plurality of peripheral agents that do not require use of ordered protocols.
(22) Another novel communication fabric technique, disclosed herein, includes a physical network topology that places source agents in one general area of an integrated circuit (IC) while destination agents may be physically arranged in other areas of the IC. Two or more communication lanes may be utilized in a particular network to couple the source and destination agents. At least some of the network switches coupled to the source agents may be cross-coupled between communication lanes to enable communication across the lanes. Network switches coupled to the destination agents may be coupled only to network switches in a same lane.
(23) A further novel communication fabric technique involves use of network interfaces with more input channels than output channels. Such network interfaces may allow for transactions to be buffered in an input channel of the network interface, thereby removing the transactions from network switches, allowing for an increased number of transactions to be routed by the network switches of a given network.
(24) For the ease of discussion, various embodiments in this disclosure are described as being implemented using one or more SoCs. It is to be understood that any disclosed SoC can also be implemented using a chiplet-based architecture. Accordingly, wherever the term “SoC” appears in this disclosure, those references are intended to also suggest embodiments in which the same functionality is implemented via a less monolithic architecture, such as via multiple chiplets, which may be included in a single package in some embodiments.
(25) On a related note, some embodiments are described herein that include more than one SoC. Such architectures are to be understood to encompass both homogeneous designs (in which each SoC includes identical or almost identical functionality) and heterogeneous designs (in which the functionality of each SoC diverges more considerably). Such disclosure also contemplates embodiments in which the functionalities of the multiple SoCs are implemented using different levels of discreteness. For example, the functionality of a first system could be implemented on a single IC, while the functionality of a second system (which could be the same or different than the first system) could be implemented using a number of co-packaged chiplets.
(26) FIG. **1** illustrates a block diagram of an embodiment of a system that uses I/O interfaces in combination with network interfaces. System **100** includes a plurality of agent circuits **120***a*-**120***h* (collectively **120**). A portion of agent circuits **120** are coupled directly to one of two network interfaces **101***a* and **101***b* (collectively **101**), while the remaining agent circuits **120** are coupled to one of network interfaces **101** via a respective one of I/O interfaces (I/O I/F) **125***a*-**125***c*

(collectively **125**). Network interfaces **101** are coupled to a communication network that includes network switching circuits (NS) **110***a* and **110***b*. Network interfaces **101** in combination with communication network **105** support transfer of data transactions **140** and **144** from respective ones of agent circuits **120**. System **100** may be, in whole or in part, a computing system, such as a desktop or laptop computer, a smartphone, a tablet computer, a wearable smart device, or the like. In some embodiments, system **100** is a single IC, such as a system-on-chip, or a multi-die chip.

(27) As shown, a first group of agent circuits (**120***d*, **120***e*, and **120***f*) are configured to transfer data transactions, including data transactions **144**, using an ordered protocol, such as a peripheral component interconnect (PCI) protocol. Such ordered protocols may specify that data transactions are transferred in an order in which they are sent from a respective agent circuit. A second group of agent circuits (**120***a*-**120***c* and **120***g*-**120***h*) are, in contrast, configured to transfer data transactions using a protocol with no enforced ordering. To support the ordered protocol for the first group of agent circuits, input/output (I/O) interfaces **125***a*, **125***b*, and **125***c* are coupled to respectively to agent circuits **120***d*, **120***e*, and **120***f*, and are configured to enforce the ordered protocol for the data transactions sent by the first group of agent circuits, including data transactions **144**.

(28) Communication network **105** includes network switching circuits **110***a* and **110***b*. As illustrated, network switching circuit **110***a* is coupled to at least network switching circuit **110***b*, and each may be further coupled to one or more additional network switching circuits in communication network **105**.

(29) Network interface **101***a*, as shown, is coupled to agent circuits **120***a*-**120***c*, to I/O interface **125***a*, and to network switching circuit **110***a*. Network interface **101***a* may be configured to transfer data transactions (including data transactions **140**) between agent circuits **120***a*-**120***c* and network switching circuit **110***a*, as well as transfer data transactions between I/O interface **125***a* and network switching circuit **110***a*. Similarly, network interface **101***b* is coupled agent circuits **120***g* and **120***h*, to I/O interfaces **125***b* and **125***c*, and to network switching circuit **110***b*. Network interface **101***b* may be configured to transfer data transactions between agent circuits **120***g* and **120***h* and network switching circuit **110***a*, as well as transfer data transactions (including data transactions **144**) between I/O interfaces **125***b* and **125***c* and network switching circuit **110***a*.

(30) As illustrated, agent circuit **120***a* transfers data transactions **140** to network interface **101***a* in a first order: data transaction **140***a*, followed by data transaction **140***b*, and then data transaction **140***c*. Since network interface **101***a* is not required to enforce an ordered protocol, data transactions **140** may be sent to network switching circuit **110***a* in any suitable order. For example, each of data transactions **140** may be directed to a different destination agent circuit, such as different memory circuits. Accordingly, network interface **101***a* may send the data transactions **140** as resources are available to receive a given transaction. Each of the memory circuits may include a respective request queue and may not be available to receive a given transaction until an entry in the respective request queue is available. Accordingly, data transactions **140** may be sent in a different order than they were received. As shown, network interface sends data transaction **140***c* first, **140***a* second and **140***b* third. In other cases, data transactions may be reordered based on their respective transaction type. For example, data transactions **140***a* and **140***b* may be write requests for a given memory circuit, while data transaction **140***c* is a read request for the same memory circuit. A read request may have a higher priority than a write request and, in some cases, may also be fulfilled faster than a write request. Network interface **101***a* may reorder data transactions **140** for a variety of reasons, some of which may increase bandwidth for communication network **105**, and/or reduce power consumed by communication network **105** and/or destination agent circuits.

(31) Agent circuit **120***e*, as shown, transfers data transactions **144** to I/O interface **125***b* in a first order: data transaction **144***a*, followed by data transaction **144***b*, and then data transaction **144***c*. I/O interface **125***b* is configured to follow the same order for data transactions **144**. In some embodiments, I/O interface **125***b* may forward each of data transactions **144** to network interface **101***b*, one at a time, waiting for an acknowledgement or other form of response before sending a

subsequent one of data transactions **144**. For example, I/O interface **125***b* may send data transaction **144***a* to network interface **101***b* and wait for a response to be sent via network interface **101***b* before sending data transaction **144***b*. In such a manner, the ordered processing of data transactions **144** may be ensured but may also be slow compared to other techniques. In other embodiments, I/O interface **125***b* may transfer data transactions **144** to network interface **101***b* in the received order. Network interface **101***b*, however, may not enforce the same order and may forward data transactions **144** in a different order. I/O interface **125***b*, however, may then receive responses associated with data transactions **144** and buffer the received responses until a response to the first transaction, data transaction **144***a*, is received, which may then be forwarded to agent circuit **120***e*. I/O interface **125***b* may continue buffering responses until a response to data transaction **144***b* is received, and so on. In such a manner, I/O interface **125***b* may process transactions, from the view of agent circuit **120***e*, in the original order.

(32) It is noted that system **100**, as illustrated in FIG. **1**, is merely an example. System **100** has been simplified to highlight features relevant to this disclosure. Elements not used to describe the details of the disclosed concepts have been omitted. For example, system **100** may include various circuits that are not illustrated, such as one or more memory circuits, clock generator circuits, power management circuits, and the like. Only one communication network is illustrated. In other embodiments, a communication fabric with any suitable number of networks may be included. In various embodiments, circuits of system **100** may be implemented using any suitable combination of sequential and combinatorial logic circuits. In addition, register and/or memory circuits, such as static random-access memory (SRAM) may be used in these circuits to temporarily hold information such as instructions, data, address values, and the like.

(33) In FIG. **1**, a communication network utilizing network interfaces and I/O interfaces is disclosed. Such network interfaces and I/O interfaces may perform additional processing not disclosed in the description of FIG. **1**. An example of translating between different data formats is depicted in FIG. **2**.

(34) Moving to FIG. **2**, a portion of the block diagram of system **100** of FIG. **1** is shown. Illustrated in FIG. **2** are agents **120***a*-**120***d*, network interface **101***a*, network switching circuit **110***a*, and I/O interface **125***a*. This portion of system **100** is used to depict how data formats of transactions may be modified as they progress from a source agent to the communication network.

(35) As illustrated, agent circuit **120***d* is configured to send data transaction **242***a* to I/O interface **125***a* using a first protocol (protocol 0). Agent circuit **120***a* is configured to send data transaction **240***a* to network interface **101***a* using a second protocol (protocol 1), that is different than protocol 0. In various embodiments, different protocols may include different data formats as well as different rules for sending and receiving data packets. Protocol 1, for example, may specify a data packet with a destination address in a first bit field, a packet size in a second bit field, a source/owner process identifier in a third bit field and then one or more data words as specified by the packet size. Rules for sending and receiving packets using protocol 1 may include receiving an acknowledgement from the destination agent as well as one or more rules governing packet priority versus other transactions using the same protocol. Protocol 0 may specify similar information in data packets, but the bit fields may be arranged in a different order. In addition, protocol 0 may specify an ordered processing of transactions and, therefore, may include a packet number in an additional bit field usable to determine this order.

(36) I/O interface **125***a*, as shown, is configured to send data transaction **242***a* received from agent circuit **120***d* to network interface **101***a* using protocol 1. Since network interface **101***a* is coupled to multiple different agents, circuit designers may desire to simplify a design of network interface **101** to support a limited number of protocols. If a particular agent circuit, e.g., agent circuit **120***d*, does not support one of the limited number of protocols, then a respective I/O interface such as I/O interface **125***a* may be included between agent circuits without support for the limited protocols (e.g., agent circuit **120***d*) and network interface **101***a*.

(37) Accordingly, I/O interface **125***a* is configured to translate data transaction **242***a* from protocol 0 to protocol 1, thereby generating data transaction **242***b*. Such a translation may include rearranging data in various bit fields of protocol 0 into corresponding bit fields of protocol 1, removing extraneous information not included in protocol 1, adding information required by protocol 1 that is not supported in protocol 0, and the like. I/O interface **125***a* sends data transaction **242***b* to network interface **101***a* after the translation.
(38) Network interface **101***a* may, in turn, be configured to translate received data transactions **240***a* and **242***b* into a third protocol (protocol 2) that is supported by network switching circuit **110***a*, as well as other network switching circuits in communication network **105**. Accordingly, these translations of data transactions **240***a* and **242***b* into protocol 2 may generate transactions **240***b* and **242***c*, respectively. Like the translation from protocol 0 to protocol 1, the translation from protocol 1 to protocol 2 may include various combinations of rearranging, adding, and removing data among a plurality of bit fields in a data packet.
(39) Although transactions are illustrated as moving from agent circuits **120***a* and **120***d* to communication network **105**, the opposite may also be implemented. Accordingly, network interface **101***a* may be configured to translate transactions coming from network switching circuit **110***a* from protocol 2 to protocol 1 prior to sending to any of agent circuits **120***a*-**120***c* or I/O interface **125***a*. Likewise, I/O interface **125***a* may translate transactions received from network interface **101***a* from protocol 1 to protocol 0 prior to sending the transactions to agent circuit **120***d*.
(40) It is noted that the system of FIG. **2** is simplified for clarity. In other embodiments, any suitable number of agent circuits, I/O interfaces, network interfaces network switching circuits and the like may be included in other embodiments. Although network interface **101***a* is described as translating between protocols 1 and 2, other network interfaces may translate between other protocols. For example, network interface **101***b* in FIG. **1** may use a fourth protocol for data packets sent to and from agent circuits g and h and/or I/O interfaces **125***b* and **125***c*.
(41) FIG. **2** depicts how different protocols may be handled for different agent circuits. In the description of FIG. **1**, protocols that enforce an ordered processing of transactions were disclosed. FIG. **3** depicts such an embodiment.
(42) Continuing to FIG. **3**, a same portion of the block diagram of system **100** of FIG. **1** is illustrated. Depicted in FIG. **3** are agents **120***a*-**120***d*, network interface **101***a*, network switching circuit **110***a*, and I/O interface **125***a*. This portion of system **100** is used to depict how an order of transactions may be adjusted as they progress from a source agent to the communication network via an I/O interface.
(43) As illustrated, agent circuit **120***d* is configured to send, in a first order, a series of data transactions (transactions **342***a*-**342***c*) to I/O interface **125***a*. Agent circuit **120***d* may be configured to send and receive data transactions **342***a*-**342***c* in a particular order. For example, agent circuit **120***d* may be configured to utilize a PCI protocol in which data transactions are expected to be processed in a same order as they are sent. The PCI protocol may also expect acknowledgements or other types of responses associated with sent data transactions. As illustrated, agent circuit **120***d* sends transaction **342***a*, followed by transaction **342***b*, and then transaction **342***c*. This is referred to herein as the original order.
(44) I/O interface **125***a*, as shown, is configured to send data transactions **342***a*-**342***c* to network interface **101***a* in the first order, e.g., **342***a*, **342***b*, and then **342***c*. I/O interface **125***a*, in some embodiments, may be configured to use the same original order to send data transactions **342***a*-**342***c* to network interface **101***a*. Network interface **101***a*, however, may be configured to send data transactions **342***a*-**342***c* to communication network **105** in a second order, different than the first, original order. For example, one or more of data transactions **342***a*-**342***c* may be directed to a different destination agent than the other transactions. One of these different destination agents may have a full request queue and not be able to receive new transactions for a period of time, while other ones of the destination agents may have available resources and therefore be able to receive

new transactions. Network interface **101***a* may, therefore, send transaction **342***c* first due to the associated destination agent being available to receive transactions, while transactions **342***a* and **342***b* are delayed due to unavailable resources for their respective destination agents.

(45) Network interface **101***a* may use other criteria as well to determine an order for sending data transactions **342***a*-**342***c*. For example, each of data transactions **342***a*-**342***c* may have a respective priority assigned, such bulk transaction or real time transaction. A bulk transaction may correspond to a standard or lowest priority for data transactions, while a real time transaction may correspond to a high priority in which the transaction needs to be processed with a minimal amount of latency. In some embodiments, agent circuit **120***d* may include a plurality of processor cores, each processor core capable of executing a different software process. Priorities may be associated with these different processes and, in turn, be assigned to ones of the transactions issued by the respective process.

(46) As illustrated, network interface **101***a* receives, from network switching circuit **110***a*, responses to each of data transactions **342***a*-**342***c* in a third order that is different from the original or second orders. As depicted, response **344***b*, corresponding to transaction **342***b*, is received first, followed by response **344***c* (corresponding to transaction **342***c*) and then response **344***a* (corresponding to transaction **342***a*). The response may be received in the third order due to relative amounts of time each of the destination agents takes to receive and process the transactions. In addition, network traffic between network switching circuit **110***a* and network switching circuits coupled to each of the destination agents may delay sending the transactions as well as receiving the responses. If a given destination agent is at a far end of communication network **105**, then transactions and responses may travel through tens, or even hundreds, of network switching circuits.

(47) Network interface **101***a*, as shown, is configured to send responses **344***a*-**344***c* to I/O interface **125***a* in the received third order. For example, network interface **101***a* may forward each response as it is received. I/O interface **125***a*, on the other hand, is configured to send responses **344***a*-**344***c* to agent circuit **120***d* in the original first order. To send responses **344***a*-**344***c* to agent circuit **120***d* in the original order, I/O interface **125***a* is configured to buffer one or more of responses **344***a*-**344***c* as they are received from network interface **101***a* in the third order. I/O interface **125***a*, for example, may include a buffer circuit capable of storing multiple responses. Accordingly, I/O interface **125***a* may buffer response **344***b* and response **344***c* until response **344***a* has been received. After response **344***a* is received, it is forwarded to agent circuit **120***d*, followed by response **344***b* and then response **344***c*. Agent circuit **120***d*, therefore, receives responses **344***a*-**344***c* in the original order, and I/O interface complies with the ordered protocol even though network interface **101***a* does not comply with the ordered protocol.

(48) Agent circuit **120***a*, as depicted, is configured to send a series of data transactions (transactions **340***a*-**340***c*) to network interface **101***a*, in a given order. Transaction **340***a* is first, followed by transaction **340***b*, and then transaction **340***c* last. Network interface **101***a* may use one or more of the criteria described above to determine an order for sending data transactions **340***a*-**340***c* to network switching circuit **110***a*. As shown, the order is transaction **340***a* first, transaction **340***c* second, and transaction **340***b* third.

(49) Responses to these transactions are received at a later point in time, in another order. Responses **341***a*-**341***c* are received in the order, response **341***c* first, response **341***a* second and response **341***b* third. As agent circuit **120***a* does not enforce an ordered protocol, network interface **101***a* may forward the responses to agent circuit **120***a* in the same order in which they are received.

(50) It is noted that the embodiment of FIG. **3** is merely an example. A limited number of elements are shown to illustrate the disclosed concepts. In other embodiments, any suitable number of each element may be included in other embodiments. Although I/O interface **125***a* is described as maintaining the original order, in some embodiments, I/O interface **125***a* may reorder transactions from agent circuit **120***d*, for example, moving a real time transaction in front of one or more bulk

transactions. I/O interface **125***a* may, in some embodiments, still send responses to agent circuit **120***d* in the original order.

(51) In system **100** of FIGS. **1-3**, network interface circuits are shown coupled to a mix of agent circuits with and without I/O interface circuits. In some embodiments, particular groups of agent circuits may be coupled to a same network interface circuit. FIG. **4** illustrates such an embodiment.

(52) Proceeding to FIG. **4**, a block diagram of an embodiment of a system that groups particular sets of agent circuits with common network interface circuits. System **400** includes agent circuits **420***a*-**420***h* (collectively **420**). Agent circuits **420***a*-**420***d* are coupled directly to network interface **401***a*, while agent circuits **420***e*-**420***h* are coupled to network interface **401***b* via respective ones of I/O interfaces (I/O I/F) **425***a*-**425***d* (collectively **425**). Network interfaces **401***a* and **401***b* are coupled to communication network **405** via network switching circuits (NS) **410***a* and **410***b*, respectively. In a similar manner as system **100**, system **400** may be included, in whole or in part, in a computing system, such as a desktop or laptop computer, a smartphone, a tablet computer, a wearable smart device, or the like. In some embodiments, system **400** is a single IC, such as a system-on-chip, or is a multi-die chip.

(53) System **400** illustrates a different arrangement of agent circuits to network interfaces. As illustrated, agent circuits **420***e*-**420***h* may be configured to transfer data transactions using an ordered protocol such as described above. Accordingly, I/O interfaces **425***a*-**425***d* are coupled to respective ones of agent circuits **420***e*-**420***h*, as well as to network interface **401***b*. I/O interfaces **425***a*-**425***d* are configured to enforce the ordered protocol when agent circuits **420***e*-**420***h* send and receive transactions and/or responses via network interface **401***b*. Network interface **401***b* is further coupled to network switching circuit **410***b*.

(54) In a manner as previously disclosed, any of agent circuits **420***e*-**420***h* may send transactions in a first order, using a first protocol, and respective ones of I/O interfaces **425***a*-**425***d* may translate the transactions into a second protocol, and forward the translated transactions to network interface **401***b* in the first order. Network interface **401***b* may then translate the transactions from the second protocol to a third protocol, and may reorder the transaction prior to transferring them to other network switching circuits to respective destination agents. Received responses to these transactions may be forwarded to the respective I/O interfaces **425***a*-**425***d* as they are received, regardless of the original order of the transactions. I/O interfaces **425***a*-**425***d*, however, may buffer received responses that do not confirm to the first order. These responses may be forwarded to the respective agent circuit **420***e*-**420***h* when a response to a first transaction in the first order is received.

(55) Network interface **401***a*, in contrast, is coupled directly to agent circuits **420***a*-**420***d* and to network switching circuit **410***a*. As described above, any of agent circuits **420***a*-**420***d* may send transactions in a first order, using the second protocol, and network interface **401***a* may translate the transactions from the second protocol to the third protocol, and may reorder the transactions prior to transferring them to other network switching circuits to respective destination agents. Received responses to these transactions may be forwarded to the respective agent circuit **420***a*-**420***d* as they are received, regardless of the original order of the transactions.

(56) By grouping agent circuits with common attributes to a common network interface, a given network interface may be optimized for the common attributes. Respective ones of the I/O interface circuits, however, may still be used for each agent circuit that follows an ordered protocol. This may allow a single or a few I/O interface circuit designs to be created and reused across a plurality of IC designs. The individual I/O interface circuits may also support scalability as more or fewer agent circuits that follow the ordered protocol may be used in various IC designs.

(57) It is noted that system **400**, as illustrated in FIG. **4**, is merely an example to demonstrate disclosed techniques. System **400** has been simplified to highlight features relevant to this disclosure. Although a single communication network is illustrated, a communication fabric with any suitable number of networks may be included in other embodiments. As described for system

100, circuits of system **400** may be implemented using any suitable combination of sequential and combinatorial logic circuits, as well as register and/or memory circuits.
(58) The circuits and techniques described above in regards to FIGS. **1-4** may be performed using a variety of methods. Two methods associated with transferring transactions via a communication network are described below in regard to FIGS. **5-6**.
(59) Turning now to FIG. **5**, a flow diagram for an embodiment of a method for transferring transactions issued by two different agent circuits is illustrated. Method **500** may be performed by any of the systems disclosed herein, such as systems **100** and **400** of FIGS. **1-4**. Method **500** is described below using system **100** of FIG. **1** as an example. References to elements in FIG. **1** are included as non-limiting examples.
(60) At **510**, method **500** begins by transferring, by a first agent to an I/O interface, a first group of data transactions using an ordered protocol. For example, agent circuit **120***e* may send data transactions **144** to I/O interface **125***b* in a first order. Agent circuit **120***e* may be configured to expect data transactions **144** to be processed in the first order which, as shown, corresponds to data transaction **144***a* first, data transaction **144***b* second, and data transaction **144***c* third. Enforcement of the first order may include expecting responses corresponding to data transactions **144** to follow the first order.
(61) Method **500** continues at **520** by transferring, by a second agent to a network interface circuit, a second group of data transactions using an unforced-order protocol. For example, agent circuit **120***a* may send data transactions **140** to network interface **101***a* in a first order corresponding to data transaction **140***a* first, data transaction **140***b* second, and data transaction **140***c* third. Agent circuit **120***a*, unlike agent circuit **120***e*, may be configured to allow processing of data transactions **140** occur in any suitable order. Responses to ones of data transactions **140** may be received by agent circuit **120***a* in any order.
(62) At **530** method **500** proceeds with transferring, by the I/O interface to the network interface circuit, the first group of data transactions using the ordered protocol. As shown in FIG. **1**, I/O interface **125***b* may forward data transactions **144** to network interface **101***b* using the first order. In some embodiments, I/O interface **125***b* may delay sending a subsequent one of data transactions **144** until a previously sent transaction **144** has been forwarded by network interface **101***b*. Such a technique may enable I/O interface **125***b* to enforce the first order. In other embodiments, I/O interface **125***b* may not delay between transfers of data transactions **144**, which may allow network interface **101***b* to reorder the data transactions **144**.
(63) Method **500** proceeds at **540** with transferring, by the network interface circuit to via a communication fabric, the first and second groups of data transactions using an order based on respective destination availability. Network interfaces **101***a* and **101***b* may determine availability of destination agents for each of data transactions **140** and **144**, respectively. Network interfaces **101***a* and **101***b* may further determine relative priorities between respective ones of data transactions **140** and **144**. Using such availability and priority information, network interface **101***a* may determine to transfer data transaction **140***c* before data transactions **140***a* and **140***b*. In some cases, network interface **101***b* may determine that the first order may be maintained and, therefore, may transfer data transactions **144** using the first order. In other embodiments, network interface **101***b* may reorder data transactions **144** due to, e.g., availability of resources of one or more of the destination agents.
(64) Use of I/O interfaces with agent circuits that enforce ordered transactions may allow enforcement to be limited to only agent circuits that require ordered processing of transactions. Other solutions may place the enforcement of ordered transactions in network interfaces that are coupled to pluralities of agent circuits, some of which may not confirm to an ordered protocol. Forcing ordered processing onto agent circuits that do not confirm to such protocols may increase latencies for these agent circuits to complete transactions, thereby potentially reducing a performance bandwidth of the system.

(65) It is noted that the method of FIG. **5** includes blocks **510-540**. Method **500** may end in block **540** or may repeat some or all blocks of the method. For example, method **500** may repeat blocks **510** and/or **520** repeatedly to transfer additional transactions. In some embodiments, different instances of method **500** may be performed concurrently in system **100**. For example, agent circuit **120***a* and network interface **101***b* may perform portions of one instance of method **500** while agent circuit **120***e*, I/O interface **125***b*, and network interface **101***b* perform portions of a different instance of method **500**.
(66) Proceeding now to FIG. **6**, a flow diagram for an embodiment of a method for receiving responses to previously issued transactions issued by an agent circuit that uses an ordered protocol is illustrated. Like method **500**, method **600** may be performed by any of the systems disclosed herein including, for example, systems **100** and **400** of FIGS. **1-4**. Method **600** is described below using system **100** of FIG. **3** as an example. References to elements in FIG. **3** are included as non-limiting examples. Operations of method **600** may occur after an instance of method **500** has been performed.
(67) Method **600** begins in **610** by receiving, by the network interface circuit via the communication fabric, responses to first and second groups of data transactions in an order based on respective destination response times. As shown in FIG. **3**, for example, agent circuit **120***a* issues data transactions **340***a*-**340***c* and agent circuit **120***d* issues data transactions **342***a*-**342***c*. Each agent circuit sends their respective transactions in the order a-b-c. As these transactions are forwarded to their respective destination agents, the transactions may be reordered. At a later point in time, network interface **101***a* receives responses to each of the issued transactions. Network interface **101***a* receives responses **341***a*-**341***c* in the order c-a-b, and receives responses **344***a*-**344***c* in the order b-c-a.
(68) At **620**, method **600** proceeds by transferring, by the network interface circuit to the I/O interface, the responses to the first group of data transactions based on the received order. For example, network interface **101***a* sends responses **344***a*-**344***c* to I/O interface **125***a* in the received order, e.g., b-c-a.
(69) Method **600** continues at **630** with transferring, by the I/O interface to the first agent, the responses to the first group of data transactions using the ordered protocol. As depicted in FIG. **3**, I/O interface **125***a* forwards responses **344***a*-**344***c* to agent circuit **120***d* in the original order, e.g., a-b-c. To accomplish this, I/O interface **125***a* may buffer responses **344***b* and **344***c* until response **344***a* is received. After response **344***a* is received, it may be forwarded to agent circuit **120***d*, followed by response **344***b* and then response **344***c*, thus presenting responses **344***a*-**344***c* to agent circuit **120***d* in the original issued order.
(70) At **640**, method **600** proceeds with transferring, by the network interface circuit to the second agent, the responses to the second group of data transactions based on the received order. As described above, agent circuit **120***a* may not be configured to enforce an ordered protocol. Accordingly, network interface **101***a* may forward the responses to agent circuit **120***a* in the same order in which they are received, e.g., response **341***c* first, response **341***a* second, and response **341***b* third.
(71) It is noted that method **600** includes blocks **610-640**. Method **600** may end in block **640** or may repeat some or all blocks of the method. For example, method **600** may return to block **610** to receive additional responses to other previously issued transactions. Various instances of methods **500** and **600** may be performed concurrently.
(72) In FIGS. **1-6**, systems and techniques are disclosed for using I/O interfaces in communication networks that include a plurality of agent circuits, some of which may enforce ordered protocols and some of which may not. Other techniques may be implemented for increasing bandwidth in a communication network. One such technique is disclosed in FIGS. **7-10**.
(73) Moving to FIG. **7**, a block diagram of an embodiment of a system that includes a multi-lane communication network is depicted. As illustrated, system **700** includes two sets of network

switching circuits. Network switching circuits **710**_a_-**710**_g_ are included in communication lane **707**_a_. Network switching circuits **712**_a_-**712**_g_ are included in communication lane **707**_b_. System **700** further includes a plurality of agent circuits **720**_a_-**720**_f_ and a plurality of memory circuits **725**_a_-**725**_h_. Communication lanes **707**_a_ and **707**_b_ are included in communication network **705**. In a similar manner as described for system **100** in FIG. **1**, system **700** may be, in whole or in part, a laptop or desktop computer, a smartphone, a tablet computer, a wearable smart device, or the like. In some embodiments, system **700** may be a single IC, such as a system-on-chip, or may be a multi-die chip.

(74) As illustrated, agent circuits **720**_a_-**720**_f_ may be configured to initiate memory transactions, as well as to receive memory transactions. Agent circuits **720**_a_-**720**_f_ may be any circuit (e.g., CPU, GPU, neural processing engine, peripheral, memory controller, etc.) that may source and/or receive communications on communication network **705**. Memory circuits **725**_a_-**725**_h_ may be configured to respond to memory transactions from agent circuits **720**_a_-**720**_f_. For example, memory circuits **725**_a_-**725**_h_ may include any suitable combinations of volatile and non-volatile memory cells including, for example, static random-access memory (SRAM), dynamic random-access memory (DRAM), flash memory, a hard drive, register files, and the like. Given ones of memory circuits **725**_a_-**725**_h_ may be configured to process a given memory transaction based on an address accessed by the given memory transaction, including read transactions to access values stored in the given memory circuits **725**_a_-**725**_h_ and write transactions to store values supplied by respective ones of agent circuits **720**_a_-**720**_f_.

(75) Communication network **705**, as shown, includes network switching circuits **710**_a_-**710**_c_ coupled to agent circuits **720**_a_-**720**_c_, respectively, and network switching circuits **712**_a_-**712**_c_ coupled to agent circuits **720**_a_-**720**_c_, respectively. Communication network **705** also includes network switching circuits **710**_d_-**710**_g_ coupled to memory circuits **725**_a_-**725**_d_ and network switching circuits **712**_d_-**712**_g_ coupled to memory circuits **725**_e_-**725**_h_. Communication network **705** is configured to transfer memory transactions via communication lanes **707**_a_ and **707**_b_.

(76) As depicted, communication lane **707**_a_ includes network switching circuits **710**_a_-**710**_g_ and is therefore coupled to a proper subset of agent circuits **720**_a_-**720**_f_, e.g., agent circuits **720**_a_-**720**_c_, as well as to a proper subset of memory circuits **725**_a_-**725**_h_, e.g., memory circuits **725**_a_-**725**_d_. In a similar manner, communication lane **707**_b_ includes network switching circuits **712**_a_-**712**_g_ and is therefore coupled to the proper subset of agent circuits **720**_d_-**720**_f_, as well as to the proper subset of memory circuits **725**_e_-**725**_h_. It is noted that the proper subsets of agent circuits and memory circuits are mutually exclusive between communication lanes **707**_a_ and **707**_b_.

(77) Network switching circuits **710**_a_-**710**_c_ in communication lane **707**_a_ are coupled to respective ones of network switching circuits **712**_a_-**712**_c_ in communication lane **707**_b_. In some embodiments, network switching circuits **710**_a_-**710**_c_ and network switching circuits **712**_a_-**712**_c_ may be coupled to form a mesh sub-network between communication lanes **707**_a_ and **707**_b_. In contrast, network switching circuits **710**_d_-**710**_g_ in communication lane **707**_a_ are isolated from network switching circuits **712**_a_-**712**_g_ in communication lane **707**_b_. Similarly, network switching circuits **712**_d_-**712**_g_ in communication lane **707**_b_ are isolated from network switching circuits **710**_a_-**710**_g_ in communication lane **707**_a_. As used herein, "isolated from" refers to a lack of a direct link from the isolated network switching circuits in one communication lane to network switching circuits in a different communication lane, and vice versa.

(78) In some embodiments, agent circuits **720**_a_-**720**_f_ may be physically located in one particular region of an IC die, such as at one end of the die. Accordingly, network switching circuits **710**_a_-**710**_c_ and **712**_a_-**712**_c_ may be placed in the same region. In addition, these network switching circuits **710**_a_-**710**_c_ and **712**_a_-**712**_c_ may be coupled to form a mesh network, enabling transfer of transactions between any group of agent circuits **720**_a_-**720**_f_. A close physical proximity between these agent circuits may reduce latency of communications between the agent circuits. In some cases, agent circuits **720**_a_-**720**_f_ may frequently communicate among one another and, therefore,

reducing latency (as compared to having agent circuits spread farther from one another across the IC die) may improve performance of system **700**.
(79) Memory circuits **725***a*-**725***h* may, in contrast, not communicate between one another. Instead, memory circuits **725***a*-**725***h* may more frequently communicate with various ones of agent circuits **720***a*-**720***f*. Accordingly, memory circuits **725***a*-**725***h* may be placed farther from one another across other regions of the IC die. Use of the multiple communication lanes, including communication lanes **707***a* and **707***b*, may help to mitigate traffic congestion if multiple agent circuits **720***a*-**720***f* are accessing respective ones of memory circuits **725***a*-**725***h*. Use of communication lanes **707***a* and **707***b* may help to distribute memory transactions across different sets of network switching circuits **710***a*-**710***g* and **712***a*-**712***g*, thereby reducing latency due to multiple transactions having to pass through common ones of network switching circuits **710***a*-**710***g* and **712***a*-**712***g*.
(80) Despite the division of communication network **705** into communication lanes **707***a* and **707***b*, any of agent circuits **720***a*-**720***f* may be capable of sending and receiving memory transactions to any of memory circuits **725***a*-**725***h*. For example, agent circuit **720***b*, coupled to network switching circuit **710***b* in communication lane **707***a*, may be configured to initiate a particular memory transaction for memory circuit **725***h* coupled to network switching circuit **712***g* in communication lane **707***b*. To send the memory transaction, agent circuit **720***b* may transfer the particular memory transaction to network switching circuit **710***b*. Network switching circuit **710***b*, in turn, may be configured to transmit the particular memory transaction to network switching circuit **712***b* in communication lane **707***b*. As illustrated, network switching circuit **710***b* may be configured to transmit the particular memory transaction directly to network switching circuit **712***b* without using an intermediate network switching circuit.
(81) Network switching circuit **712***b* may be configured to transmit the particular memory transaction to network switching circuit **712***g* in communication lane **707***b*. To access memory circuit **725***h*, network switching circuit **712***b* may be further configured to send the particular memory transaction via one or more intermediate network switching circuits in communication lane **707***b* (e.g., network switching circuits **712***c* and **712***f* in the illustrated example.
(82) It is noted that system **700**, as shown in FIG. **7**, is merely an example. Like system **100** in FIG. **1**, system **700** has been simplified to highlight features relevant to this disclosure and omit elements that are not relevant. For example, system **700** may include various circuits that are not illustrated, such as clock generator circuits, power management circuits, and the like. Although the disclosed communication lanes are described as being included in one communication network, in other embodiments, a communication fabric with any suitable number of networks may be included. The circuits of system **700** may be implemented using any suitable combination of sequential and combinatorial logic circuits. Memory circuits may be implemented suitable types of memory cells, such as SRAM, DRAM, flash, and the like.
(83) In FIG. **7**, a single communication network, e.g., on a single IC die is disclosed. Some systems may implement a multi-die solution, each die including at least one respective communication network. An example of a multi-die embodiment is depicted in FIG. **8**.
(84) Moving to FIG. **8**, a block diagram of an embodiment of a system that includes two IC dies, each including a respective multi-lane communication network is illustrated. As shown, system **800** includes SoCs **801***a* and **801***b*. In various embodiments, SoCs **801***a* and **801***b* may be packaged separately and coupled together via a circuit board, coupled together within a common package, or coupled together via any other suitable manner. SoC **801***a* includes network switching circuits **810***a*-**810***g* and **812***a*-**812***e*, agent circuits **820***a*-**820***f* and memory circuits **825***a*-**825***f*. Communication lanes **807***a* and **807***b* are included in communication network **805***a*. SoC **801***b* includes network switching circuits **810***h*-**810***k* and **812***f*-**812***j*, agent circuits **820***g*-**820***j* and memory circuits **825***g*-**825***k*. Communication lanes **807***c* and **807***d* are included in communication network **805***b*. In a similar manner as described in regards to, for example, system **700** in FIG. **7**, system **800** may be included in a laptop or desktop computer, a smartphone, a tablet computer, a

wearable smart device, or the like.

(85) As illustrated, SoC **801***b* includes agent circuit **820***h* which is a die-to-die interface circuit. In addition, agent circuit **820***b* in SoC **801***a* includes a die-to-die interface circuit configured to transfer memory transactions between SoCs **801***a* and **801***b*. The die-to-die interfaces in agent circuits **820***b* and **820***h* may enable SoCs **801***a* and **801***b* to communicate and process transactions such that an operating system executed on SoC **801***a* and/or **801***b* may perform as if system **800** were a single IC.

(86) For example, agent circuit **820***b* is coupled to network switching circuit **810***b* in communication lane **807***a* of SoC **801***a*. Agent circuit **820***f* is coupled to network switching circuit **812***c* in communication lane **807***b* and may be configured to initiate a particular memory transaction for memory circuit **825***j* in SoC **801***b*. To transfer the particular memory transaction to memory circuit **825***j*, agent circuit **820***f* may be configured to send the particular memory transaction to network switching circuit **810***c* which, in turn, may be configured to transfer the particular memory transaction to network switching circuit **810***b* via either network switching circuit **810***c* or **812***b*, both of which are coupled to network switching circuit **810***b*.

(87) Network switching circuit **810***b* may, as illustrated, be configured to transfer the particular memory transaction to agent circuit **820***b*. Using the die-to-die interfaces, agent circuit **820***b* transfers the particular memory transaction to agent circuit **820***h* in SoC **801***b*. Agent circuit **820***h* may be configured to transfer the particular memory transaction network switching circuit **812***i* via network switching circuits **810***i*, **812***f*, and **812***h*. The particular memory transaction may then be delivered to memory circuit **825***j* via network switching circuit **812***j*.

(88) Use of multiple communication lanes may enable SoCs **801***a* and **801***b* to be coupled to one another via a single die-to-die interface on each SoC. Use of the multiple communication lanes may help to avoid traffic congestion once transactions have crossed over the die-to-die interface into the destination SoC. Furthermore, agent circuits **820***a*-**820***j* may, as described for system **700** in FIG. **7**, be physically placed in respective regions of each SoC, near the respective die-to-die interface. Such placement may help to reduce transaction latency from source agent circuits to the die-to-die interfaces.

(89) It is noted that that the system of FIG. **8**, is an example for demonstrating the disclosed techniques. System **800** has been simplified to show elements relevant to this demonstration. Although one communication network is shown for each SoC, in other embodiments, a communication fabric with any suitable number of networks may be included.

(90) In FIGS. **1-8**, various embodiments of a communication network are disclosed. These communications networks may be a single network or a network fabric that includes a plurality of networks. Whether a single network or a fabric of networks, each network may be implemented using any suitable type of network topology, with a given network fabric including networks of varying structure. FIGS. **9** and **10** illustrate two examples of network topologies.

(91) Turning to FIG. **9**, a block diagram of an embodiment of a network using a ring topology to couple a plurality of agent circuits is shown. In the example of FIG. **9**, the ring is formed from network switching circuits **914**AA-**914**AH. Agent circuit **910**A is coupled to network switching circuit **914**AA; agent circuit **910**B is coupled to network switching circuit **914**AB; and agent circuit **910**C is coupled to network switching circuit **914**AE.

(92) As shown, a "network switching circuit," or simply "network switch" is a circuit that is configured to receive communications on a network and forward the communications on the network in the direction of the destination of the communication. For example, a communication sourced by a processor may be transmitted to a memory controller that controls the memory that is mapped to the address of the communication. At each network switch, the communication may be transmitted forward toward the memory controller. If the communication is a read, the memory controller may communicate the data back to the source and each network switching circuit may forward the data on the network toward the source. In an embodiment, the network may support a

plurality of virtual channels. The network switching circuit may employ resources dedicated to each virtual channel (e.g., buffers) so that communications on the virtual channels may remain logically independent. The network switching circuit may also employ arbitration circuitry to select among buffered communications to forward on the network. Virtual channels may be channels that physically share a network, but which are logically independent on the network (e.g., communications in one virtual channel do not block progress of communications on another virtual channel).

(93) In a ring topology, each network switching circuit **914**AA-**914**AH may be connected to two other network switching circuits **914**AA-**914**AH, and the switches form a ring such that any network switching circuit **914**AA-**914**AH may reach any other network switching circuit in the ring by transmitting a communication on the ring in the direction of the other network switch. A given communication may pass through one or more intermediate network switching circuits in the ring to reach the targeted network switching circuit. When a given network switching circuit **914**AA-**914**AH receives a communication from an adjacent network switching circuit **914**AA-**914**AH on the ring, the given network switching circuit may examine the communication to determine if an agent circuit **910**A-**910**C to which the given network switching circuit is coupled is the destination of the communication. If so, the given network switching circuit may terminate the communication and forward the communication to the agent. If not, the given network switching circuit may forward the communication to the next network switching circuit on the ring (e.g., the other network switching circuit **914**AA-**914**AH that is adjacent to the given network switching circuit and is not the adjacent network switching circuit from which the given network switching circuit received the communication). As used herein, an "adjacent network switch" to a given network switching circuit may be a network switching circuit to which the given network switching circuit may directly transmit a communication, without the communication traveling through any intermediate network switching circuits.

(94) The example of FIG. **9** is one example of a ring network topology. As illustrated, any pair of adjacent network switching circuits may communicate in both directions, as indicated by the arrows. In some embodiments, however, a ring network may allow communication in only one direction, e.g., only clockwise or only counterclockwise. Such embodiments may be used, for example, to simplify design of each of the network switching circuits.

(95) Proceeding to FIG. **10**, a block diagram of one embodiment of a network using a mesh topology to couple agent circuits **1010**A-**1010**P is illustrated. As shown in FIG. **10**, network **1000** may include network switching circuits **1014**AA-**1014**AH. Network switching circuits **1014**AA-**1014**AH are coupled to two or more other network switching circuits. For example, network switching circuit **1014**AA is coupled to network switching circuits **1014**AB and **1014**AE; network switching circuit **1014**AB is coupled to network switching circuits **1014**AA, **1014**AF, and **1014**AC; etc. as illustrated in FIG. **10**. Thus, individual network switching circuits in a mesh network may be coupled to a different number of other network switching circuits. Furthermore, while network **1000** has a relatively symmetrical structure, other mesh networks may be asymmetrical, for example, depending on the various traffic patterns that are expected to be prevalent on the network. At each network switching circuit **1014**AA-**1014**AH, one or more attributes of a received communication may be used to determine the adjacent network switching circuit **1014**AA-**1014**AH to which the receiving network switching circuit **1014**AA-**1014**AH will transmit the communication (unless an agent circuit **1010**A-**1010**P to which the receiving network switching circuit **1014**AA-**1014**AH is coupled is the destination of the communication, in which case the receiving network switching circuit **1014**AA-**1014**AH may terminate the communication on network **1000** and provide it to the destination agent circuit **1010**A-**1010**P). For example, in an embodiment, network switching circuits **1014**AA-**1014**AH may be programmed at system initialization to route communications based on various attributes.

(96) In an embodiment, communications may be routed based on the destination agent. The

routings may be configured to transport the communications through the fewest number of network switching circuits (the **37** shortest path) between the source and destination agent that may be supported in the mesh topology. Alternatively, different communications for a given source agent to a given destination agent may take different paths through the mesh. For example, latency-sensitive communications may be transmitted over a shorter path while less critical communications may take a different path to avoid consuming bandwidth on the short path, where the different path may be less heavily loaded during use, for example. Additionally, a path may change between two particular network switching circuits for different communications at different times. For example, one or more intermediate network switching circuits in a first path used to transmit a first communication may experience heavy traffic volume when a second communication is sent at a later time. To avoid delays that may result from the heavy traffic, the second communication may be routed via a second path that avoids the heavy traffic.

(97) FIG. **10** may be an example of a partially-connected mesh: at least some communications may pass through one or more intermediate network switching circuits in the mesh. A fully-connected mesh may have a connection from each network switching circuit to each other network switch, and thus any communication may be transmitted without traversing any intermediate network switching circuits. Any level of interconnectedness may be used in various embodiments.

(98) In the descriptions of FIGS. **1-6**, network interface circuits are described in relation to network switching circuits. Communication between network switching circuits and network interfaces may be implemented in a variety of fashions. An example of how a network switching circuit may exchange transactions with a network interface is shown in FIG. **11**.

(99) Moving now to FIG. **11**, a block diagram of an embodiment of system with an agent circuit, a network interface and a network switch is shown. As illustrated, system **1100** includes agent circuit **1120** coupled to network interface (I/F) **1101** which, in turn, is coupled to network switching circuit **1110***a*. Network switching circuit **1110***a* is further coupled to network switching circuits **1110***b* and **1110***c*. Network switching circuits **1110***a***-1110***c* are included in communication network **1105**. In some embodiments, agent circuit **1120**, network interface **1101**, and network switching circuits **1110***a***-1110***c* may correspond to similarly named and numbered elements shown in FIGS. **1-4**.

(100) As illustrated, agent circuit **1120** may be configured to transfer data transactions to and from communication network **1105** via network interface **1101**. Communication network **1105**, as disclosed, includes network switching circuits **1110***a***-1110***c* for transferring data transactions between agent circuit **1120** and other agent circuits coupled to other network switching circuits (not shown) in communication network **1105**.

(101) Network interface **1101**, as shown, includes a plurality of input channels (Rx CH **1145***a* and **1145***b*) and one or more output channels (Tx CH **1143**). Network interface **1101** may be configured to receive a first plurality of data transactions (data transactions **1140***a* and **1140***b*) from network switching circuit **1110***a* via Rx CH **1145***a* and **1145***b*. In some embodiments, network interface **1101** is further configured to receive data transactions **1140***a* and **1140***b* from network switching circuit **1110***a* concurrently. Network interface **1101** may be further configured to send a second plurality of data transactions (data transactions **1140***c* and **1140***d*) to network switching circuit **1110***a* via Tx CH **1143**. As depicted, network interface **1101** may be further configured to send data transactions **1140***c* and **1140***d* to network switching circuit **1110***a* serially. To receive data transactions **1140***a* and **1140***b* (as well as subsequent data transactions) concurrently, network interface **1101** may, in some embodiments, include respective transaction buffers coupled to Rx CHs **1145***a* and **1145***b*. In addition, network interface **1101** may include arbitration circuit (arbiter) **1147** to select a data transaction from a given one of the transaction buffers. Arbitration circuit **1147** may use any suitable arbitration scheme to select between Rx CHs **1145***a* and **1145***b*, such as least recently used, round robin, credits, number of buffered transactions, and the like. In some embodiments, Tx CH **1143** may be coupled to a respective buffer circuit while, in other embodiments, agent circuit **1120** may send data transactions to Tx CH **1143** one at a time.

(102) In some embodiments, agent circuit **1120** may be configured to receive data transactions **1140** at a first data rate. For example, agent circuit **1120** may be a graphics processing unit or a display capable of consuming data transactions at the first rate to support displaying frames of a video at a particular frame rate and resolution. In such embodiments, failure to receive the data transactions at the first data rate may result in the video playback stalling or glitching. To send data transactions **1140** to agent circuit **1120** at the first data rate, network switching circuit may be configured to enter a first performance state to serially send data transactions **1140** to agent circuit **1120** at the first data rate, and to enter a second performance state to concurrently send data transactions **1140** to agent circuit **1120** at the first data rate. For example, to serially transfer data transactions **1140** from network switching circuit **1110***a* to network interface **1101** at the first rate, communication network **1105** may operate in a performance mode that enables data transfers at the first data rate. Clock frequencies in such a performance mode must be high enough to support the serial data rate. Sending data transactions concurrently, however, may allow use of a slower frequency (e.g., one-half of the frequency of the first performance mode) since two data transactions may be transferred in parallel. Accordingly, the second performance state may use less power than the first performance state.
(103) To avoid congestion at network switching circuit **1110***a* when agent circuit **1120** is sending data transactions **1140***c* and **1140***d*, network switching circuit **1110***a* may include virtual channel queue (VQ) **1155***a* that is configured to receive data transactions **1140***c* and **1140***d* from network interface **1101**. Network switching circuit **1110***a* may further include virtual channel queue (VQ) **1155***b* that is configured to receive a different plurality of data transactions from another network switching circuit in communication network **1105** (e.g., network switching circuit **1110***b*). Accordingly, network switching circuit **1110***a* may be capable of receiving data transactions from network switching circuit **1110***b* and network interface **1101** concurrently, and placing them into virtual channel queues **1155***b* and **1155***a*, respectively. Arbitration circuit (arbiter) **1157** may be configured to use any suitable arbitration algorithm (such as those disclosed above) to select a data transaction from either of virtual channel queues **1155***a* or **1155***b*.
(104) By using a first number of input channels that is greater than a second number of output channels, network interface may be capable of pulling data transactions out of communication network **1105** more quickly than if data transactions were received serially. This may relieve congestion in communication network **1105**, particularly in network switching circuit **1110***a*, thereby increasing a bandwidth for transferring data transaction. In addition, if data traffic in communication network **1105** is not heavy, then communication network **1105** may be capable of running in a lower performance mode, thereby reducing power consumption while maintaining a desired data rate for transferring data transactions to agent circuit **1120**.
(105) It is noted that that system **1100** of FIG. **11**, is merely an example. System **1100** has been simplified to show elements relevant to this demonstration. For example, a single agent circuit is shown coupled to the network interface. In other embodiments, a plurality of agent circuits may be coupled to a single network interface. Although one output channel and two input channels are shown for network interface **1101**, in other embodiments, any suitable number of input and output channels may be included. A number of input channels, however, may be greater than a number of output channels.
(106) To summarize, various embodiments are disclosed including an apparatus that may comprise one or more first agent circuits that are configured to transfer data transactions using an ordered protocol, as well as one or more second agent circuits that are configured to transfer data transactions using a protocol with no enforced ordering. This apparatus may also include one or more input/output (I/O) interfaces coupled to respective ones of the first agent circuits, and are configured to enforce the ordered protocol. Furthermore, the apparatus may include a communication network including a plurality of network switching circuits. A particular one of the plurality of network switching circuits may be coupled to at least one other network switching

circuit of the plurality of network switching circuits. The apparatus may also include a network interface circuit, coupled the second agent circuits, to the I/O interfaces, and to the particular network switching circuit. This network interface circuit may be configured to transfer data transactions between the second agent circuits and the particular network switching circuit, and to transfer data transactions between the I/O interfaces and the particular network switching circuit.
(107) In a further example, a particular first agent circuit may be configured to send data transactions to a respective I/O interface using a first protocol. A particular second agent circuit may be configured to send data transactions to the network interface circuit using a second protocol, different than the first protocol. In another example, the respective I/O interface is configured to send data transactions received from the particular first agent circuit to the network interface circuit using the second protocol.
(108) In an example, a particular first agent circuit may be configured to send, in a first order, a series of data transactions to a respective I/O interface. The respective I/O interface may be configured to send the series of data transactions to the network interface circuit in the first order. The network interface circuit may be configured to send the series of data transactions to the communication network in a second order, different than the first order.
(109) Another apparatus may comprise a particular integrated circuit including a plurality of agent circuits configured to initiate memory transactions. The particular integrated circuit may also include a plurality of memory circuits configured to respond to the memory transactions. A given memory circuit may be configured to process a given memory transaction based on an address accessed by the given memory transaction. The particular integrated circuit may further include a communication network, including one or more first network switching circuits coupled to a first portion of the plurality of agent circuits, and one or more second network switching circuits coupled to a second portion of the plurality of memory circuits. The communication network may be configured to transfer memory transactions via one of two communication lanes. A first communication lane may include a first proper subset of the first and second network switching circuits, and a second communication lane may include a second proper subset of the first and second network switching circuits. The first and second proper subsets may be mutually exclusive. At least one of the first network switching circuits in the first communication lane may be coupled to a respective one of the first network switching circuits in the second communication lane. The second network switching circuits in the first communication lane may be isolated from network switching circuits in the second communication lane.
(110) In a further example, a plurality of the first network switching circuits in the first communication lane and a plurality of the first network switching circuits in the second communication lane may be coupled to form a mesh sub-network between the first and second communication lanes. In an example, a particular agent, coupled to a particular first network switching circuit in the first communication lane, may be configured to initiate a particular memory transaction for a particular memory circuit coupled to a particular second network switching circuit in the second communication lane.
(111) In another example, the particular first network switching circuit may be configured to transmit the particular memory transaction to a different first network switching circuit in the second communication lane. The different first network switching circuit may be configured to transmit the particular memory transaction to the particular second network switching circuit in the second communication lane.
(112) Another example of an apparatus may comprise an agent circuit configured to transfer data transactions, and a communication network including a plurality of network switching circuits. A particular one of the plurality of network switching circuits may be coupled to at least one other network switching circuit of the plurality of network switching circuits. The apparatus may also include a network interface circuit, coupled to the agent circuit and to the particular network switching circuit, and may include a plurality of input channels and one or more output channels. A

first number of the input channels may be greater than a second number of the one or more output channels. The network interface circuit may be configured to receive a first plurality of data transactions from the particular network switching circuit via the plurality of input channels, and to send a second plurality of data transactions to the particular network switching circuit via the one or more output channels.

(113) In a further example, the network interface circuit may be further configured to receive the first plurality of data transactions from the particular network switching circuit concurrently. In another example, the network interface circuit may be further configured to send the second plurality of data transactions to the particular network switching circuit serially.

(114) In an example, the agent circuit may be configured to receive data transactions at a first data rate. To send data transactions to the agent circuit at the first data rate, the particular network switching circuit may be configured to enter a first performance state to serially send the data transactions to the agent circuit at the first data rate, and enter a second performance state to concurrently send the data transactions to the agent circuit at the first data rate. The second performance state may use less power than the first performance state.

(115) The circuits and techniques described above in regards to FIG. **11** may be performed using a variety of methods. An example method associated with transferring transactions via a network interface is described below in regard to FIG. **12**.

(116) Proceeding now to FIG. **12**, a flow diagram for an embodiment of a method for transferring data transactions to and from a network interface is illustrated. Method **1200** may be used in conjunction with any of the systems disclosed herein, for example, system **1100** in FIG. **11**. Method **1200** is described below using system **1100** of FIG. **11** as an example. References to elements in FIG. **12** are included as non-limiting examples. As depicted, method **1200** may be performed concurrently with the previously described methods **500** and **600**.

(117) Method **1200** begins in **1210** with a network interface circuit receiving a first plurality of data transactions from the particular network switching circuit via a plurality of input channels. For example, network interface **1101** may receive data transactions **1140***a* and **1140***b* concurrently from network switching circuit **1110***a*, and may place received data transactions **1140***a* and **1140***b* into respective buffers coupled to each of Rx CHs **1145***a* and **1145***b*. Data transactions **1140***a* and **1140***b* may each be received at a first data rate per input channel.

(118) At **1220**, method **1200** proceeds with the network interface circuit sending the first plurality of data transactions to an agent circuit coupled to the network interface circuit using a second data rate that is higher than the first data rate. For example, arbitration circuit **1147** may be used to select either data transaction **1140***a* from Rx CH **1145***a* or data transaction **1140***b* from Rx CH **1145***b*. As described above, any suitable arbitration technique may be utilized. Arbitration circuit **1147** may then forward the selected data transaction to agent circuit **1120** at the second data rate, higher than the first data rate. Use of the lower data rate to transfer data transactions **1140** from network switching circuit **1110***a* to network interface **1101** may allow communication network **1105** to operate in a reduced power state while maintaining a desired data rate for providing data transactions to agent circuit **1120**.

(119) Method **1200** continues at **1230** with the network interface circuit receiving a second plurality of data transactions from the agent circuit using the second data rate. Agent circuit **1120**, for example, may generate data transactions **1140***c* and **1140***d* to be sent to an agent circuit (not shown in FIG. **11**) that is also coupled to communication network **1105**. Agent circuit **1120** may send data transactions **1140***c* and **1140***d* to network interface **1101**. In some embodiments, network interface **1101** may receive data transactions **1140***a* and **1140***b* one at a time. In other embodiments, Tx CH **1143** may include, or be coupled to, a buffer that allows multiple data transactions to be received and queued for sending.

(120) At **1240**, method **1200** may continue with the network interface circuit sending the second plurality of data transactions to the particular network switching circuit via one or more output

channels. For example, network interface **1101** sends data transactions **1140***a* and **1140***b* to network switching circuit **1110***a* via Tx CH **1143**. Since network interface **1101**, as shown, has a single output channel (Tx CH **1143**), data transactions **1140***a* and **1140***b* are sent serially. As is illustrated in FIG. **11**, a first number of the input channels (Rx CH **1145**) is greater than a second number of the one or more output channels (Tx CH **1143**). In some embodiments, agent circuit **1120** may not generate as many data transactions as it receives. If, for example, agent circuit **1120** is a display, then agent circuit **1120** may receive many large data transactions corresponding to a frame of image data to display, while sending modest amounts of data transactions to, for example, reply to status requests, and/or indicate too much or too little image data is being received. By including additional input channels, data transactions may be offloaded from communication network **1105** more rapidly, thereby freeing bandwidth within communication network **1105**. Since agent circuit **1120** may not generate as many data transactions as it consumes, additional output channels may not be implemented in some embodiments.

(121) It is noted that method **1200** includes blocks **1210-1240**. Method **1200** may end in block **1240** or may repeat some or all blocks of the method. For example, method **1200** may repeat operations **1210** and **1220** to receive a large number of data transactions. Method **1200** may be performed concurrently with methods **500** and **600**. Method **1200** may also be performed concurrently with another instance of method **1200**. For example, a network interface circuit may be coupled to a plurality of agent circuits and have a plurality of input channels for two or more of these agent circuits.

(122) FIGS. **1-12** illustrate circuits and methods for a system, such as an integrated circuit, that include a variety of network interfaces and I/O interfaces for transferring transactions between agent circuits in a communication network. Any embodiment of the disclosed systems may be included in one or more of a variety of computer systems, such as a desktop computer, laptop computer, smartphone, tablet, wearable device, and the like. In some embodiments, the circuits described above may be implemented on a system-on-chip (SoC) or other type of integrated circuit, including multi-die packages. A block diagram illustrating an embodiment of system **1300** is illustrated in FIG. **13**. System **1300** may, in some embodiments, include any disclosed embodiment of systems disclosed herein, such as systems **100**, **400**, **700**, **800**, and **1100** shown in various ones of FIGS. **1-11**.

(123) In the illustrated embodiment, the system **1300** includes at least one instance of a system on chip (SoC) **1306** which may include multiple types of processor circuits, such as a central processing unit (CPU), a graphics processing unit (GPU), or otherwise, a communication fabric, and interfaces to memories and input/output devices. SoC **1306** may correspond to an instance of the SoCs disclosed herein. In various embodiments, SoC **1306** is coupled to external memory circuit **1302**, peripherals **1304**, and power supply **1308**.

(124) A power supply **1308** is also provided which supplies the supply voltages to SoC **1306** as well as one or more supply voltages to external memory circuit **1302** and/or the peripherals **1304**. In various embodiments, power supply **1308** represents a battery (e.g., a rechargeable battery in a smart phone, laptop or tablet computer, or other device). In some embodiments, more than one instance of SoC **1306** is included (and more than one external memory circuit **1302** is included as well.

(125) External memory circuit **1302** is any type of memory, such as dynamic random-access memory (DRAM), synchronous DRAM (SDRAM), double data rate (DDR, DDR2, DDR3, etc.) SDRAM (including mobile versions of the SDRAMs such as mDDR3, etc., and/or low power versions of the SDRAMs such as LPDDR2, etc.), RAMBUS DRAM (RDRAM), static RAM (SRAM), etc. In some embodiments, external memory circuit **1302** may include non-volatile memory such as flash memory, ferroelectric random-access memory (FRAM), or magnetoresistive RAM (MRAM). One or more memory devices may be coupled onto a circuit board to form memory modules such as single inline memory modules (SIMMs), dual inline memory modules

(DIMMs), etc. Alternatively, the devices may be mounted with a SoC or an integrated circuit in a chip-on-chip configuration, a package-on-package configuration, or a multi-chip module configuration.

(126) The peripherals **1304** include any desired circuitry, depending on the type of system **1300**. For example, in one embodiment, peripherals **1304** includes devices for various types of wireless communication, such as Wi-Fi, Bluetooth, cellular, global positioning system, etc. In some embodiments, the peripherals **1304** also include additional storage, including RAM storage, solid state storage, or disk storage. The peripherals **1304** include user interface devices such as a display screen, including touch display screens or multitouch display screens, keyboard or other input devices, microphones, speakers, etc.

(127) As illustrated, system **1300** is shown to have application in a wide range of areas. For example, system **1300** may be utilized as part of the chips, circuitry, components, etc., of a desktop computer **1310**, laptop computer **1320**, tablet computer **1330**, cellular or mobile phone **1340**, or television **1350** (or set-top box coupled to a television). Also illustrated is a smartwatch and health monitoring device **1360**. In some embodiments, the smartwatch may include a variety of general-purpose computing related functions. For example, the smartwatch may provide access to email, cellphone service, a user calendar, and so on. In various embodiments, a health monitoring device may be a dedicated medical device or otherwise include dedicated health related functionality. In various embodiments, the above-mentioned smartwatch may or may not include some or any health monitoring related functions. Other wearable devices **1360** are contemplated as well, such as devices worn around the neck, devices attached to hats or other headgear, devices that are implantable in the human body, eyeglasses designed to provide an augmented and/or virtual reality experience, and so on.

(128) System **1300** may further be used as part of a cloud-based service(s) **1370**. For example, the previously mentioned devices, and/or other devices, may access computing resources in the cloud (i.e., remotely located hardware and/or software resources). Still further, system **1300** may be utilized in one or more devices of a home **1380** other than those previously mentioned. For example, appliances within the home may monitor and detect conditions that warrant attention. Various devices within the home (e.g., a refrigerator, a cooling system, etc.) may monitor the status of the device and provide an alert to the homeowner (or, for example, a repair facility) should a particular event be detected. Alternatively, a thermostat may monitor the temperature in the home and may automate adjustments to a heating/cooling system based on a history of responses to various conditions by the homeowner. Also illustrated in FIG. **13** is the application of system **1300** to various modes of transportation **1390**. For example, system **1300** may be used in the control and/or entertainment systems of aircraft, trains, buses, cars for hire, private automobiles, waterborne vessels from private boats to cruise liners, scooters (for rent or owned), and so on. In various cases, system **1300** may be used to provide automated guidance (e.g., self-driving vehicles), general systems control, and otherwise.

(129) It is noted that the wide variety of potential applications for system **1300** may include a variety of performance, cost, and power consumption requirements. Accordingly, a scalable solution enabling use of one or more integrated circuits to provide a suitable combination of performance, cost, and power consumption may be beneficial. These and many other embodiments are possible and are contemplated. It is noted that the devices and applications illustrated in FIG. **13** are illustrative only and are not intended to be limiting. Other devices are possible and are contemplated.

(130) As disclosed in regards to FIG. **13**, system **1300** may include one or more integrated circuits included within a personal computer, smart phone, tablet computer, or other type of computing device. A process for designing and producing an integrated circuit using design information is presented below in FIG. **14**.

(131) FIG. **14** is a block diagram illustrating an example of a non-transitory computer-readable

storage medium that stores circuit design information, according to some embodiments. The embodiment of FIG. **14** may be utilized in a process to design and manufacture integrated circuits, for example, including one or more instances of systems (or portions thereof) **100**, **400**, **700**, **800**, and **1100** that are disclosed above. In the illustrated embodiment, semiconductor fabrication system **1420** is configured to process the design information **1415** stored on non-transitory computer-readable storage medium **1410** and fabricate integrated circuit **1430** based on the design information **1415**.
(132) Non-transitory computer-readable storage medium **1410**, may comprise any of various appropriate types of memory devices or storage devices. Non-transitory computer-readable storage medium **1410** may be an installation medium, e.g., a CD-ROM, floppy disks, or tape device; a computer system memory or random-access memory such as DRAM, DDR RAM, SRAM, EDO RAM, Rambus RAM, etc.; a non-volatile memory such as a Flash, magnetic media, e.g., a hard drive, or optical storage; registers, or other similar types of memory elements, etc. Non-transitory computer-readable storage medium **1410** may include other types of non-transitory memory as well or combinations thereof. Non-transitory computer-readable storage medium **1410** may include two or more memory mediums which may reside in different locations, e.g., in different computer systems that are connected over a network.
(133) Design information **1415** may be specified using any of various appropriate computer languages, including hardware description languages such as, without limitation: VHDL, Verilog, SystemC, SystemVerilog, RHDL, M, MyHDL, etc. Design information **1415** may be usable by semiconductor fabrication system **1420** to fabricate at least a portion of integrated circuit **1430**. The format of design information **1415** may be recognized by at least one semiconductor fabrication system, such as semiconductor fabrication system **1420**, for example. In some embodiments, design information **1415** may include a netlist that specifies elements of a cell library, as well as their connectivity. One or more cell libraries used during logic synthesis of circuits included in integrated circuit **1430** may also be included in design information **1415**. Such cell libraries may include information indicative of device or transistor level netlists, mask design data, characterization data, and the like, of cells included in the cell library.
(134) Integrated circuit **1430** may, in various embodiments, include one or more custom macrocells, such as memories, analog or mixed-signal circuits, and the like. In such cases, design information **1415** may include information related to included macrocells. Such information may include, without limitation, schematics capture database, mask design data, behavioral models, and device or transistor level netlists. As used herein, mask design data may be formatted according to graphic data system (gdsii), or any other suitable format.
(135) Semiconductor fabrication system **1420** may include any of various appropriate elements configured to fabricate integrated circuits. This may include, for example, elements for depositing semiconductor materials (e.g., on a wafer, which may include masking), removing materials, altering the shape of deposited materials, modifying materials (e.g., by doping materials or modifying dielectric constants using ultraviolet processing), etc. Semiconductor fabrication system **1420** may also be configured to perform various testing of fabricated circuits for correct operation.
(136) In various embodiments, integrated circuit **1430** is configured to operate according to a circuit design specified by design information **1415**, which may include performing any of the functionality described herein. For example, integrated circuit **1430** may include any of various elements shown or described herein. Further, integrated circuit **1430** may be configured to perform various functions described herein in conjunction with other components.
(137) As used herein, a phrase of the form "design information that specifies a design of a circuit configured to . . . " does not imply that the circuit in question must be fabricated in order for the element to be met. Rather, this phrase indicates that the design information describes a circuit that, upon being fabricated, will be configured to perform the indicated actions or will include the specified components.

(138) The present disclosure includes references to an “embodiment” or groups of “embodiments” (e.g., “some embodiments” or “various embodiments”). Embodiments are different implementations or instances of the disclosed concepts. References to “an embodiment,” “one embodiment,” “a particular embodiment,” and the like do not necessarily refer to the same embodiment. A large number of possible embodiments are contemplated, including those specifically disclosed, as well as modifications or alternatives that fall within the spirit or scope of the disclosure.
(139) This disclosure may discuss potential advantages that may arise from the disclosed embodiments. Not all implementations of these embodiments will necessarily manifest any or all of the potential advantages. Whether an advantage is realized for a particular implementation depends on many factors, some of which are outside the scope of this disclosure. In fact, there are a number of reasons why an implementation that falls within the scope of the claims might not exhibit some or all of any disclosed advantages. For example, a particular implementation might include other circuitry outside the scope of the disclosure that, in conjunction with one of the disclosed embodiments, negates or diminishes one or more the disclosed advantages. Furthermore, suboptimal design execution of a particular implementation (e.g., implementation techniques or tools) could also negate or diminish disclosed advantages. Even assuming a skilled implementation, realization of advantages may still depend upon other factors such as the environmental circumstances in which the implementation is deployed. For example, inputs supplied to a particular implementation may prevent one or more problems addressed in this disclosure from arising on a particular occasion, with the result that the benefit of its solution may not be realized. Given the existence of possible factors external to this disclosure, it is expressly intended that any potential advantages described herein are not to be construed as claim limitations that must be met to demonstrate infringement. Rather, identification of such potential advantages is intended to illustrate the type(s) of improvement available to designers having the benefit of this disclosure. That such advantages are described permissively (e.g., stating that a particular advantage “may arise”) is not intended to convey doubt about whether such advantages can in fact be realized, but rather to recognize the technical reality that realization of such advantages often depends on additional factors.
(140) Unless stated otherwise, embodiments are non-limiting. That is, the disclosed embodiments are not intended to limit the scope of claims that are drafted based on this disclosure, even where only a single example is described with respect to a particular feature. The disclosed embodiments are intended to be illustrative rather than restrictive, absent any statements in the disclosure to the contrary. The application is thus intended to permit claims covering disclosed embodiments, as well as such alternatives, modifications, and equivalents that would be apparent to a person skilled in the art having the benefit of this disclosure.
(141) For example, features in this application may be combined in any suitable manner. Accordingly, new claims may be formulated during prosecution of this application (or an application claiming priority thereto) to any such combination of features. In particular, with reference to the appended claims, features from dependent claims may be combined with those of other dependent claims where appropriate, including claims that depend from other independent claims. Similarly, features from respective independent claims may be combined where appropriate.
(142) Accordingly, while the appended dependent claims may be drafted such that each depends on a single other claim, additional dependencies are also contemplated. Any combinations of features in the dependent that are consistent with this disclosure are contemplated and may be claimed in this or another application. In short, combinations are not limited to those specifically enumerated in the appended claims.
(143) Where appropriate, it is also contemplated that claims drafted in one format or statutory type (e.g., apparatus) are intended to support corresponding claims of another format or statutory type

(e.g., method).

(144) Because this disclosure is a legal document, various terms and phrases may be subject to administrative and judicial interpretation. Public notice is hereby given that the following paragraphs, as well as definitions provided throughout the disclosure, are to be used in determining how to interpret claims that are drafted based on this disclosure.

(145) References to a singular form of an item (i.e., a noun or noun phrase preceded by “a,” “an,” or “the”) are, unless context clearly dictates otherwise, intended to mean “one or more.” Reference to “an item” in a claim thus does not, without accompanying context, preclude additional instances of the item. A “plurality” of items refers to a set of two or more of the items.

(146) The word “may” is used herein in a permissive sense (i.e., having the potential to, being able to) and not in a mandatory sense (i.e., must).

(147) The terms “comprising” and “including,” and forms thereof, are open-ended and mean “including, but not limited to.”

(148) When the term “or” is used in this disclosure with respect to a list of options, it will generally be understood to be used in the inclusive sense unless the context provides otherwise. Thus, a recitation of “x or y” is equivalent to “x or y, or both,” and thus covers 1) x but not y, 2) y but not x, and 3) both x and y. On the other hand, a phrase such as “either x or y, but not both” makes clear that “or” is being used in the exclusive sense.

(149) A recitation of “w, x, y, or z, or any combination thereof” or “at least one of . . . w, x, y, and z” is intended to cover all possibilities involving a single element up to the total number of elements in the set. For example, given the set [w, x, y, z], these phrasings cover any single element of the set (e.g., w but not x, y, or z), any two elements (e.g., w and x, but not y or z), any three elements (e.g., w, x, and y, but not z), and all four elements. The phrase “at least one of . . . w, x, y, and z” thus refers to at least one element of the set [w, x, y, z], thereby covering all possible combinations in this list of elements. This phrase is not to be interpreted to require that there is at least one instance of w, at least one instance of x, at least one instance of y, and at least one instance of z.

(150) Various “labels” may precede nouns or noun phrases in this disclosure. Unless context provides otherwise, different labels used for a feature (e.g., “first circuit,” “second circuit,” “particular circuit,” “given circuit,” etc.) refer to different instances of the feature. Additionally, the labels “first,” “second,” and “third” when applied to a feature do not imply any type of ordering (e.g., spatial, temporal, logical, etc.), unless stated otherwise.

(151) The phrase “based on” is used to describe one or more factors that affect a determination. This term does not foreclose the possibility that additional factors may affect the determination. That is, a determination may be solely based on specified factors or based on the specified factors as well as other, unspecified factors. Consider the phrase “determine A based on B.” This phrase specifies that B is a factor that is used to determine A or that affects the determination of A. This phrase does not foreclose that the determination of A may also be based on some other factor, such as C. This phrase is also intended to cover an embodiment in which A is determined based solely on B. As used herein, the phrase “based on” is synonymous with the phrase “based at least in part on.”

(152) The phrases “in response to” and “responsive to” describe one or more factors that trigger an effect. This phrase does not foreclose the possibility that additional factors may affect or otherwise trigger the effect, either jointly with the specified factors or independent from the specified factors. That is, an effect may be solely in response to those factors, or may be in response to the specified factors as well as other, unspecified factors. Consider the phrase “perform A in response to B.” This phrase specifies that B is a factor that triggers the performance of A, or that triggers a particular result for A. This phrase does not foreclose that performing A may also be in response to some other factor, such as C. This phrase also does not foreclose that performing A may be jointly in response to B and C. This phrase is also intended to cover an embodiment in which A is performed

solely in response to B. As used herein, the phrase “responsive to” is synonymous with the phrase “responsive at least in part to.” Similarly, the phrase “in response to” is synonymous with the phrase “at least in part in response to.”

(153) Within this disclosure, different entities (which may variously be referred to as “units,” “circuits,” other components, etc.) may be described or claimed as “configured” to perform one or more tasks or operations. This formulation-[entity] configured to [perform one or more tasks]-is used herein to refer to structure (i.e., something physical). More specifically, this formulation is used to indicate that this structure is arranged to perform the one or more tasks during operation. A structure can be said to be “configured to” perform some task even if the structure is not currently being operated. Thus, an entity described or recited as being “configured to” perform some task refers to something physical, such as a device, circuit, a system having a processor unit and a memory storing program instructions executable to implement the task, etc. This phrase is not used herein to refer to something intangible.

(154) In some cases, various units/circuits/components may be described herein as performing a set of task or operations. It is understood that those entities are “configured to” perform those tasks/operations, even if not specifically noted.

(155) The term “configured to” is not intended to mean “configurable to.” An unprogrammed FPGA, for example, would not be considered to be “configured to” perform a particular function. This unprogrammed FPGA may be “configurable to” perform that function, however. After appropriate programming, the FPGA may then be said to be “configured to” perform the particular function.

(156) For purposes of United States patent applications based on this disclosure, reciting in a claim that a structure is “configured to” perform one or more tasks is expressly intended not to invoke 35 U.S.C. § 112 (f) for that claim element. Should Applicant wish to invoke Section 112(f) during prosecution of a United States patent application based on this disclosure, it will recite claim elements using the “means for” [performing a function] construct.

(157) Different “circuits” may be described in this disclosure. These circuits or “circuitry” constitute hardware that includes various types of circuit elements, such as combinatorial logic, clocked storage devices (e.g., flip-flops, registers, latches, etc.), finite state machines, memory (e.g., random-access memory, embedded dynamic random-access memory), programmable logic arrays, and so on. Circuitry may be custom designed, or taken from standard libraries. In various implementations, circuitry can, as appropriate, include digital components, analog components, or a combination of both. Certain types of circuits may be commonly referred to as “units” (e.g., a decode unit, an arithmetic logic unit (ALU), functional unit, memory management unit (MMU), etc.). Such units also refer to circuits or circuitry.

(158) The disclosed circuits/units/components and other elements illustrated in the drawings and described herein thus include hardware elements such as those described in the preceding paragraph. In many instances, the internal arrangement of hardware elements within a particular circuit may be specified by describing the function of that circuit. For example, a particular “decode unit” may be described as performing the function of “processing an opcode of an instruction and routing that instruction to one or more of a plurality of functional units,” which means that the decode unit is “configured to” perform this function. This specification of function is sufficient, to those skilled in the computer arts, to connote a set of possible structures for the circuit.

(159) In various embodiments, as discussed in the preceding paragraph, circuits, units, and other elements may be defined by the functions or operations that they are configured to implement. The arrangement and such circuits/units/components with respect to each other and the manner in which they interact form a microarchitectural definition of the hardware that is ultimately manufactured in an integrated circuit or programmed into an FPGA to form a physical implementation of the microarchitectural definition. Thus, the microarchitectural definition is

recognized by those of skill in the art as structure from which many physical implementations may be derived, all of which fall into the broader structure described by the microarchitectural definition. That is, a skilled artisan presented with the microarchitectural definition supplied in accordance with this disclosure may, without undue experimentation and with the application of ordinary skill, implement the structure by coding the description of the circuits/units/components in a hardware description language (HDL) such as Verilog or VHDL. The HDL description is often expressed in a fashion that may appear to be functional. But to those of skill in the art in this field, this HDL description is the manner that is used transform the structure of a circuit, unit, or component to the next level of implementational detail. Such an HDL description may take the form of behavioral code (which is typically not synthesizable), register transfer language (RTL) code (which, in contrast to behavioral code, is typically synthesizable), or structural code (e.g., a netlist specifying logic gates and their connectivity). The HDL description may subsequently be synthesized against a library of cells designed for a given integrated circuit fabrication technology, and may be modified for timing, power, and other reasons to result in a final design database that is transmitted to a foundry to generate masks and ultimately produce the integrated circuit. Some hardware circuits or portions thereof may also be custom-designed in a schematic editor and captured into the integrated circuit design along with synthesized circuitry. The integrated circuits may include transistors and other circuit elements (e.g. passive elements such as capacitors, resistors, inductors, etc.) and interconnect between the transistors and circuit elements. Some embodiments may implement multiple integrated circuits coupled together to implement the hardware circuits, and/or discrete elements may be used in some embodiments. Alternatively, the HDL design may be synthesized to a programmable logic array such as a field programmable gate array (FPGA) and may be implemented in the FPGA. This decoupling between the design of a group of circuits and the subsequent low-level implementation of these circuits commonly results in the scenario in which the circuit or logic designer never specifies a particular set of structures for the low-level implementation beyond a description of what the circuit is configured to do, as this process is performed at a different stage of the circuit implementation process.
(160) The fact that many different low-level combinations of circuit elements may be used to implement the same specification of a circuit results in a large number of equivalent structures for that circuit. As noted, these low-level circuit implementations may vary according to changes in the fabrication technology, the foundry selected to manufacture the integrated circuit, the library of cells provided for a particular project, etc. In many cases, the choices made by different design tools or methodologies to produce these different implementations may be arbitrary.
(161) Moreover, it is common for a single implementation of a particular functional specification of a circuit to include, for a given embodiment, a large number of devices (e.g., millions of transistors). Accordingly, the sheer volume of this information makes it impractical to provide a full recitation of the low-level structure used to implement a single embodiment, let alone the vast array of equivalent possible implementations. For this reason, the present disclosure describes structure of circuits using the functional shorthand commonly employed in the industry.

Claims

1. An apparatus comprising: a plurality of agent circuits including: one or more first agent circuits configured to transfer data transactions using an ordered protocol; one or more second agent circuits configured to transfer data transactions using a protocol with no enforced ordering; one or more input/output (I/O) interfaces coupled to respective ones of the first agent circuits, and configured to enforce the ordered protocol; a communication network including a plurality of network switching circuits, wherein a particular one of the plurality of network switching circuits is coupled to at least one other network switching circuit of the plurality of network switching circuits; and a network interface circuit, coupled to the second agent circuits, to the I/O interfaces,

and to the particular network switching circuit, and configured to: transfer data transactions between the second agent circuits and the particular network switching circuit; and transfer data transactions between the I/O interfaces and the particular network switching circuit.

2. The apparatus of claim 1, wherein a particular first agent circuit is configured to send data transactions to a respective I/O interface using a first protocol; and wherein a particular second agent circuit is configured to send data transactions to the network interface circuit using a second protocol, different than the first protocol.

3. The apparatus of claim 2, wherein the respective I/O interface is configured to send data transactions received from the particular first agent circuit to the network interface circuit using the second protocol.

4. The apparatus of claim 1, wherein a particular first agent circuit is configured to send, in a first order, a series of data transactions to a respective I/O interface; wherein the respective I/O interface is configured to send the series of data transactions to the network interface circuit in the first order; and wherein the network interface circuit is configured to send the series of data transactions to the communication network in a second order, different than the first order.

5. The apparatus of claim 4, wherein the network interface circuit is configured to send responses to the series of data transactions to the respective I/O interface in a third order, corresponding to an order the responses are received from the communication network; and wherein the respective I/O interface is configured to send the responses to the series of data transactions to the particular first agent circuit in the first order.

6. The apparatus of claim 5, wherein to send the responses to the series of data transactions to the particular first agent circuit in the first order, the respective I/O interface is configured to buffer ones of the responses received from the network interface circuit in the third order.

7. The apparatus of claim 1, further comprising: one or more third agent circuits configured to transfer data transactions using the ordered protocol; one or more additional I/O interfaces coupled to respective ones of the third agent circuits, and configured to enforce the ordered protocol; and a different network interface circuit, coupled to the additional I/O interfaces, and to a different one of the plurality of network switching circuits.

8. The apparatus of claim 1, further comprising: a plurality of memory circuits; and a first communication lane and a second communication lane; wherein the plurality of network switching circuits include one or more first network switching circuits coupled to respective ones of a first portion of the plurality of agent circuits, and one or more second network switching circuits coupled to respective ones of the plurality of memory circuits; wherein the first communication lane includes a first proper subset of the first and second network switching circuits; and wherein a second communication lane includes a second proper subset of the first and second network switching circuits, wherein the first and second proper subsets are mutually exclusive.

9. The apparatus of claim 8, wherein at least one of the first network switching circuits in the first communication lane is coupled to a respective one of the first network switching circuits in the second communication lane; and wherein the second network switching circuits in the first communication lane are isolated from network switching circuits in the second communication lane.

10. The apparatus of claim 1, wherein the network interface circuit includes a plurality of input channels and one or more output channels, wherein a first number of the input channels is greater than a second number of the one or more output channels, and wherein the network interface circuit is further configured to: receive a first plurality of data transactions from the particular network switching circuit via the plurality of input channels; and send a second plurality of data transactions to the particular network switching circuit via the one or more output channels.

11. A method comprising: transferring, by a first agent circuit to an input/output (I/O) interface circuit, a first group of data transactions using an ordered protocol; transferring, by a second agent circuit to a network interface circuit, a second group of data transactions using an unforced-order

protocol; transferring, by the I/O interface circuit to the network interface circuit, the first group of data transactions using the ordered protocol; and transferring, by the network interface circuit to a communication fabric, the first and second groups of data transactions using an order based on respective destination availability.

12. The method of claim 11, further comprising receiving, by the network interface circuit from the communication fabric, responses to the first and second groups of data transactions in a received order that is based on respective destination response times.

13. The method of claim 12, further comprising: transferring, by the network interface circuit to the I/O interface circuit, the responses to the first group of data transactions based on the received order; and transferring, by the network interface circuit to the second agent circuit, the responses to the second group of data transactions based on the received order.

14. The method of claim 13, further comprising transferring, by the I/O interface circuit to the first agent circuit, the responses to the first group of data transactions using the ordered protocol.

15. The method of claim 11, further comprising: receiving, by the network interface circuit, a third group of data transactions from the communication fabric via a plurality of input channels, wherein the data transactions are received at a first data rate per input channel; sending, by the network interface circuit, the third group of data transactions to a particular agent circuit using a second data rate that is higher than the first data rate; receiving, by the network interface circuit, a fourth group of data transactions from the particular agent circuit using the second data rate; and sending, by the network interface circuit, the fourth group of data transactions to the communication fabric via one or more output channels, wherein a first number of the input channels is greater than a second number of the one or more output channels.

16. A system comprising: a plurality of agent circuits, including: a first agent circuit configured to send, in a first order, a first group of data transactions using an ordered protocol; a second agent circuit configured to send a second group of data transactions using a protocol with no enforced ordering; an input/output (I/O) interface coupled to the first agent circuit, and configured to: receive the first group of data transactions in the first order; and send the first group of data transactions using the first order; a communication network including a plurality of network switching circuits; and a network interface circuit and configured to: receive, from the I/O interface, the first group of data transactions in the first order; send, to a particular network switching circuit of the plurality of network switching circuits, the first group of data transactions with no enforced ordering; receive, from the second agent circuit, the second group of data transactions; and send the second group of data transactions to the particular network switching circuit with no enforced ordering.

17. The system of claim 16, further comprising a plurality of memory circuits coupled to a first portion of the plurality of network switching circuits, and wherein a portion of the plurality of agent circuits is coupled to a second portion of the plurality of network switching circuits; and wherein the communications network includes: a first communication lane that includes a first proper subset of the first and second portions of network switching circuits; and a second communication lane that includes a second proper subset of the first and second portions of network switching circuits, wherein the first and second proper subsets are mutually exclusive.

18. The system of claim 17, wherein at least one of the second portion of network switching circuits in the first communication lane is coupled to a respective one of the second portion of network switching circuits in the second communication lane; and wherein the first portion of network switching circuits in the first communication lane are isolated from the first and second portions of network switching circuits in the second communication lane.

19. The system of claim 16, wherein the network interface circuit is configured to: receive, from the particular network switching circuit, responses to the first and second groups of data transactions in a received order that is based on respective destination response times; and send, to the I/O interface, the responses to the first group of data transactions based on the received order;

and send, to the second agent circuit, the responses to the second group of data transactions based on the received order.

20. The system of claim 19, wherein the I/O interface is further configured to: receive, from the network interface circuit, the responses to the first group of data transactions in an order that is based on the received order; and send, to the first agent circuit, the responses to the first group of data transactions in the first order.